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Paris, August 11, 2025

European Patent Application No. 22702191.2 - 1111
Filed on January 19, 2022
In the names of: INSTITUT GUSTAVE ROUSSY / EVERIMMUNE
For: Biological marker of intestinal dysbiosis, useful for predicting the response of a cancer patient to an anti-PD1 drug

REPLY TO COMMUNICATION UNDER ARTICLE 94(3) EPC DATED 18.02.2025

Dear Sirs,

We refer to the Official Communication pursuant to Article 94(3) EPC issued by your services on February 18, 2025, relating to the above mentioned European Patent Application and to your mail relating to the extension of the time limit, dated June 18, 2025.

In response to this Communication, please find attached an amended set of claims as well as our comments below.

Amendments to the claims

In claims 1, 3, 4 and 5, the predetermined threshold has been defined as comprised between 3% and 10% of relative abundance. Support for this amendment can be found at least in claims 6 and 7 as filed.

Claims 6 and 7 claims have been amended consequently. Claim 6 is now identical to claim 7 as filed and in claim 7 now recites that the predetermined threshold corresponds to a relative abundance between 4,75 and 6.5%. Support for 4,75% is at least page 17, lines 21-22 of the application as filed.

Claim 11 has been amended to define the FMT composition and specify that the patient has been identified as having an overrepresentation of *Akkermansia* in his/her intestinal microbiota by the method of claim 3. Support for this amendment is at least in claim 3, alinea (iii).

Former claims 12 and 13 have been deleted and the following claims and dependencies have been renumbered accordingly.

Applicant considers that the claims as filed do not add any new matter and that the requirements of Article 123(2) EPC are fulfilled.

Comments

In the Communication, the Examining Division acknowledged that the claims filed upon entry into the regional phase before the EPO meet the requirements of Article 123(2) EPC. The ED then referred to the International Preliminary Report on Patentability (IPRP) and noted that the deficiencies mentioned in that report give rise to objections under the corresponding provisions of the EPC.

The objections are addressed below.

1. Inventive step of independent claims 1 and 3 and 11 over the combination of D1 (WO 2018/115519 A1) and D2 (Dibyendu Dutta et al., 2020)

In the Written Opinion, the Examiner considered that independent claims 1 and 3 and 13 (now claim 11) lacked inventive step compared to D1 and D2.

We respectfully disagree.

Please find below our analysis of D1 and D2, and our explanations regarding inventive step of the claimed invention.

- *Teachings of D1 (WO 2018/115519 A1)*

D1 is a prior patent application by the same team of inventors as the present application.

As exactly noted by the Examiner, D1 discloses an *in vitro* theranostic method of determining if a cancer patient is likely to be a good responder to an anti-PD1/PD-L1/PD-L2 Ab-based therapy, comprising assessing, in a feces sample from said patient, the relative abundance of microorganisms including *Akkermansia muciniphila*, and comparing the relative abundance of said organism to a predetermined threshold, wherein over-representation of *Akkermansia muciniphila* is indicative that the patient is likely to be a good responder to the anti-PD1/PD-L1/PD-L2 Ab-based therapy.

D1 thus teaches that when *Akkermansia muciniphila* is absent or present in an insufficient amount in a patient's gut microbiota, the patient is likely not to respond to an anti-PD1/PD-L1/PD-L2 Ab-based therapy.

D1 does not teach that *Akkermansia muciniphila* might have a negative effect on the response to such a treatment when its abundance is too high.

- *Teachings of D2 (Dibyendu Dutta et al., 2020)*

D2 is a review article about the role of gut microbiota in cancer development and therapy. Regarding *Akkermansia*, D2 refers to several studies showing that *A. muciniphila* improves intestinal barrier function and cancer treatment outcomes, including an article by Routy et al.

(ref. 92), written by the authors of D1 and disclosing the same results as D1 (in D2, see at least page 10, right-hand column, to page 11, first paragraph).

D2 also refers to two publications (ref. 131 and ref. 132) which are said to show that an overgrowth or over-abundance of *A. muciniphila* may be detrimental (see page 11, left-hand column, second §).

Ref. 131 (Desai *et al.*, *Cell* 2016) and ref. 132 (Rosean *et al.*, *Cancer Res.* 2019) are attached and referred to in what follows as D4 and D5, respectively.

According to the authors of D2, D4 teaches that:

> *"*A. muciniphila *degrades mucin. Its abundance may paradoxically induce damage to the mucosal barrier, promote intestinal permeability, and facilitate translocation of endotoxin resulting in negative pathological outcomes. Abundance of* A. muciniphila *was shown to promote epithelial access following mucus degradation and trigger lethal colitis by Citrobacter rodentium in gnotobiotic mice"*

A skilled person reading D2 and willing to understand the meaning of this sentence would obviously have also read D4 and found that the above sentence is a biased shortcut of the teachings of D4.

D4 reports the results of a study in which the authors tried to understand the functional interactions between dietary fiber (known to be beneficial), the gut microbiota and the colonic mucus barrier. They showed that during dietary fiber deficiency, the gut microbiota resorts to host-secreted mucus glycoproteins as a nutrient source, leading to erosion of the colonic mucus barrier. Regarding the role of commensal bacteria on *Citrobacter rodentium* infection, D4 says:

> *"[d]ietary fiber deprivation, together with a fiber-deprived, mucus-eroding microbiota, promotes greater epithelial access and lethal colitis by the mucosal pathogen, Citrobacter rodentium"* (see abstract),
>
> *"[these] data illustrate that a fiber-deprived microbiota has profound effects on the susceptibility to a gastrointestinal pathogen via reduction of this barrier"* (page 11 of D4, last §),

and the authors of D4 conclude that:

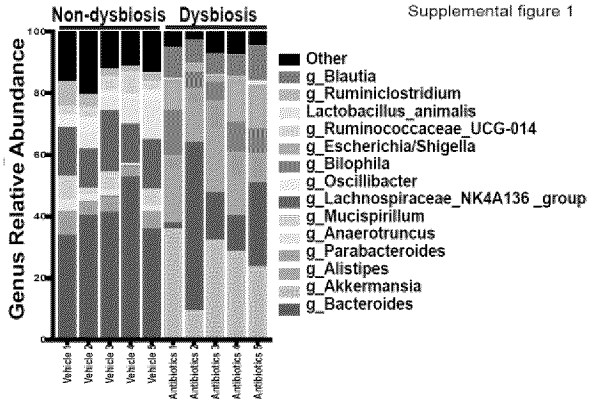
> *"efforts to find the optimal combinations of natural or prebiotic fiber polysaccharides and the minimum intake required to restore the integrity and resilience of the colonic mucus layer should be paramount"* (last sentence of the discussion, page 12)

From the above, a skilled person reading D2 would also have read D4 and understood that the statement in D2 allegedly based on D4 is flawed. What is taught by D4 is that **in absence of fiber in the regimen**, mucin-degrading commensal bacteria (such as *A. muciniphila*) would erode the colonic mucus barrier, with potential harmful consequences. The cause of this is the lack of fiber, **not the amount** of *A. muciniphila* in the host's gut.

D2 also refers to D5 as allegedly teaching that *"[a]dditional negative effects of A. muciniphila enrichment was observed in a hormone receptor-positive breast cancer mice model"* (page 11, left-hand column, second § of D2).

D5 reports a study on the role of commensal dysbiosis as a host-intrinsic factor associated with metastatic dissemination in hormone receptor-positive (HR^+) breast cancer. The authors of D5 used a mouse model of HR^+ mammary cancer, in which they induced dysbiosis by directly targeting gut microbes using non-absorbable antibiotics or by fecal microbiota transplantation of dysbiotic cecal contents. By doing so, the authors demonstrated the direct impact of gut dysbiosis on mammary tumor dissemination and identified dysbiosis as a pre-existing, host-intrinsic regulator of tissue inflammation, myeloid recruitment, fibrosis, and dissemination of tumor cells in HR^+ breast cancer.

However, contrary to what is said in D2, the authors of D5 do not say that the observed negative effects are due to *A. muciniphila* enrichment. Rather, they attribute the effects to dysbiosis in general. In fact, D5 does not cite *A. muciniphila* even once in the text of the article! The only mention of the *Akkermansia* genus in D5 is in Supplemental Figure 1, reproduced below:



As shown in this figure, *Akkermansia* is not the sole genus that increases in dysbiotic mice, and nothing in D5 suggests that the negative effects of dysbiosis are attributable to this genus in particular.

Again, a skilled person reading D2 and would also have read D5 and understood that the alleged negative effects *A. muciniphila* enrichment in hormone receptor-positive breast cancer were in fact the result of dysbiosis and **could not be attributed to *A. muciniphila*** without further experiments.

From the above, **a skilled person would obviously have disregarded the statements in D2 regarding alleged negative effects of *A. muciniphila***.

We also note that the authors of D2 say:

> "*The threshold dose needed for A. muciniphila to enhance cancer treatment response without inducing negative effects will, therefore, need to be established before any isolated supplementation or enrichment is attempted to improve the outcome of cancer therapy.*" (page 11, left-hand column, end of the second paragraph).

Instead of supplementation with *Akkermansia*, the authors of D2 advise to administer a 32 kDa protein with pili-like structure called AMUC_1100, expressed at the surface of *A. muciniphila*, and which showed beneficial effects in several mice models:

> *"Given the potential benefits and risks associated with A. muciniphila supplementation in cancer therapy, AMUC_1100 may be a safer alternative to using intact organisms".* (page 11, left-hand column, end of the third paragraph).

As another option, D2 suggests to use *A. muciniphila* derived extracellular vesicles (AmEVs):

> *"Instead of recombinant AMUC_1100, A. muciniphila derived extracellular vesicles (AmEVs) may be another option"* (page 11, paragraph bridging the two columns)

- *Could-would approach*

We agree with the Examiner that D1 can be considered as closest prior art (CPA).

We also agree with the Examiner that the method of claim 1 differs from the teachings of D1 at least in that in addition, patients with an abundance of *A. muciniphila* above a higher predetermined threshold are identified as poor responders to an immune checkpoint inhibitor (ICI)-based therapy.

The Examiner then considered that starting from D1, the objective technical problem solved by the invention is to provide an alternative method.

We respectfully disagree and submit that the objective technical problem is to provide a method for more accurately identifying patients likely to respond to an ICI-based therapy. Indeed, patients having a relative abundance above a certain threshold (with no upper limit) were considered as good responders in D1, whereas the inventors now demonstrated that above a superior threshold, the patients are likely not to respond to an ICI-based therapy.

The present invention teaches that the response of these patients can be improved by administering a bacterial compensation with a polymicrobial consortium and/or through fecal microbial transplant (FMT) from a healthy individual or from a cancer patient who successfully responded to the ICI-based therapy (point (iii) of claim 3 and claim 11).

To clarify the situation, we amended claim 1 to recite that the predetermined threshold for interpreting the relative abundance of bacteria of the *Akkermansia* genus in a patient's sample is between 3% and 10%.

We respectfully submit that a skilled person starting from D1 and willing to solve the objective technical problem, would not have found in D2 the necessary information to arrive at the claimed invention, for several reasons:

(i) the skilled person would not have understood from D2 that an overrepresentation of *Akkermansia* could be harmful as such

As explained above, a skilled person reading D2 would have been surprised by the unusual statements regarding the detrimental effects of an overgrowth or over-abundance of *A. muciniphila* written in page 11, left-hand column, second § and would have read the two publications (D4 and D5) allegedly showing these effects. By doing so, the skilled person would have recognized that

the authors of D2 made a wrong interpretation of these documents, and disregarded these statements.

In fact, a skilled person would most probably have completely disregarded D2 because the errors in the interpretation of D4 and D5 discredited its whole content

(ii) D2 teaches away from the claimed invention

As also recalled above, D2 teaches to administer either a 32 kDa protein with pili-like structure called AMUC_1100, expressed at the surface of *A. muciniphila* or *A. muciniphila* derived extracellular vesicles (AmEVs) to all the patients, because the threshold dose needed for *A. muciniphila* to enhance cancer treatment response without inducing negative effects was unknown and needed research to be determined.

Hence, a skilled person who would have considered D2 despite its deficiencies would have found no indication regarding a threshold above which *Akkermansia* would have negative effects, and would have been motivated to develop supplements not comprising live bacteria instead of those indicated in claim 3.

From the above, independent claims 1, 3 and 11 are inventive over the cited prior art and fulfil the requirements of Article 56 EPC.

2. Novelty of claim 4 over D2 (Dibyendu Dutta et al., 2020) and D3 (Weir et al., 2013)

In the Written Opinion, the Examiner considered that the subject-matter of independent claim 4 is anticipated by D2-D3, because these documents allegedly disclose that the presence of *A. muciniphila* above a threshold is indicative of dysbiosis, "*and hence, presence of this microorganism below this threshold is indicative that there is no dysbiosis*".

We agree with the Examiner that D2 and D3 disclose that *A. muciniphila* can be over-represented in dysbiosis. This is not a surprise and not the subject-matter of Claim 4. However, contrary to the Examiner's statement, the cited prior art does not at all describe that "*presence of this microorganism below this threshold is indicative that there is no dysbiosis*".

Indeed, nothing in these documents indicate that when bacteria of the *Akkermansia* genus (i) are present and (ii) not over-represented, this is enough to say that there is no dysbiosis, *i.e.*, that there is not one or several other bacterial species that are over-represented, leading to a poor diversity.

This is exactly what the inventors found, as explained at least page 20, lines 9-15, page 39, lines 36-37 and illustrated by Figure 7A:

> *"The inventors surprisingly demonstrated that the intestinal residence of bacteria of the Akkermansia genus, such as Akkermansia muciniphila and Akkermansia SGB9228, is an indirect marker of richness of the gut ecosystem, as shown by the association of Akkermansia muciniphila at a relative abundance within the 75th percentile with the alpha diversity (Shannon diversity index) of the stools, so that the level of Akkermansia muciniphila and/or Akkermansia SGB9228 and/or the Akkermansia genus can be measured to quickly and easily identify gut dysbiosis."* (page 20, lines 9-15)

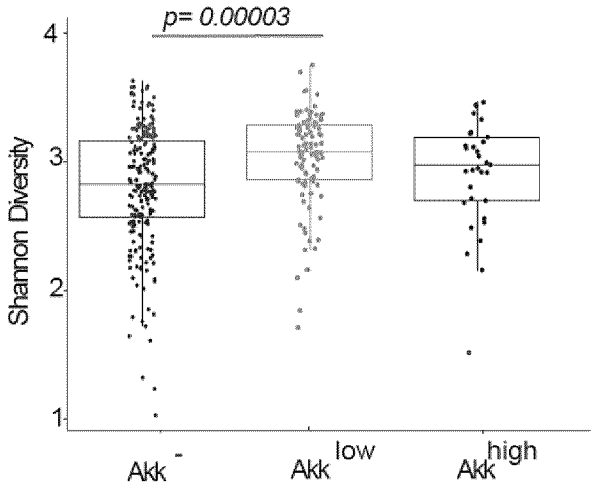


Figure 7A

We note that for ruling out dysbiosis, it is not enough that the relative abundance of *Akkermansia* be below the predetermined threshold. Indeed, as shown in the first column of Figure 7A, the absence of *Akkermansia* is associated with a low level of diversity, itself a hallmark of dysbiosis.

To clarify the situation, claim 4 has been amended as follows:

> *4. A method for determining if an individual has an intestinal microbiota dysbiosis, comprising measuring, in a sample from said individual, the relative abundance of bacteria of the Akkermansia genus, wherein the presence of bacteria of the Akkermansia genus, with a relative abundance below a predetermined threshold comprised between 3 and 10%, is indicative that there is no intestinal microbiota dysbiosis.*

The claims as amended now recites the two conditions regarding bacteria of the *Akkermansia* genus: (i) they must be present and (ii) their relative abundance must be below the predetermined threshold. In addition, the threshold has been defined as comprised between 3 and 10% of relative abundancy.

For the reasons explained above, this claim fulfils the requirements of Article 54 EPC.

3. Clarity (Article 84 EPC)

In the Written Opinion, the Examiner requested that the predetermined threshold be defined in the claims, and not as an abundance above that of a certain percentile. In response, we defined this threshold as being comprised between 3% and 10% of relative abundance.

The Examiner also requested that the fecal microbial composition of claim 12 (now claim 11) be defined, which is now the case.

Applicant considers that the attached set of claims fulfils the requirements of Article 84 EPC.

Conclusion

Applicant trusts that the above comments and amendments will clarify the situation and that a favourable issue can now be foreseen for this patent application.

Should nevertheless the Examining Division have some further minor points to discuss, it is respectfully requested to contact the undersigned Representative over the telephone and, if necessary, oral proceedings are requested in accordance with Article 116 EPC.

Respectfully yours,



Véronique MARCADÉ
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Encl.: Amended set of claims (with and without annotations)
D4: Desai *et al.*, *Cell* 2016
D5: Rosean *et al.*, *Cancer Res.* 2019 and its Supplemental Figure 1